

Assignment # 03: Functions and Data Structure (List)

Objective:

To help you understand:

- The use and implementation of functions in programming and the execution mechanism of recursive function.
- To familiarize with the list Data Structure and its built-in functions in Python

Part 1: Function

1. Write a function named "Greetings" that takes user's name and print greeting.
Output: *Welcome to SMIT training center, Ahsan*
2. Write a function that takes a number as argument and check if a given number is positive, negative or zero.
3. **Write a function to take two numbers as arguments and return the larger number.**
4. Write a function to take three numbers as argument and return the largest number.
5. **Write a function to take user's age as argument and return the message from the function whether he is minor, adult or senior citizen:**
 - a. **Minor age is less than 18.**
 - b. **Adult age is greater than 18 and less than 60**
 - c. **Senior citizen age is greater than 60**
6. Write a function to take integer as argument and check if it is even or odd.
7. Write a function to take number as argument and return the square of that number.
8. Write a function to compute the area and circumference of the circle and return the computed results.
9. Write a function to take user's score as argument and determine if they pass or fail (pass if score is above 60, otherwise fails.)
10. Write a function that evaluates if an input number is prime.
11. Write a function to compute factorial of a given number using recursion technique.

Part-2: Data Structures (List)

1. Create a list to store student data i.e. *name, age, course and is_attending*. Display each element of list using *for loop*.
List = ["Tahir", 44, "AI and Data Science", True]
2. Write a code to separate strings, numbers and Boolean data from the above list into separate lists.
3. **Create a list ["apple", "raspberry", "pineapple", "cherry"].**

- a. How can you check if apple is present in the list and get the index of the element (if present)
 - b. Now replace the raspberry and pineapple with orange.
 - c. Insert "apricot" at index 2.
 - d. Extend the resulting list with items "car", "bike", "aeroplane".
4. Write a code to get first and second best scores from the list:
Scores_list = [40, 89, 90, 89, 23, 90, 50]
5. Create a list [32,54,66,11,77,10,90]
 - a. Remove items from the list with values less than 20.
 - b. Sort the list in ascending and descending order.
 - c. Now, compute the average value of the list items.
6. From the given list:
Gadgets = ["Mobile", "Laptop", 10.0, "Marker", 3, 10, "Speakers", "Camera", 310.25]
 - a. Create separate list of string and numbers,
 - b. Sort the string list in ascending and descending order
 - c. Sort the string list by length of elements in the list
 - d. Sort the numbers list in ascending and descending order